RATIONAL EQUIVALENCE ON A PRODUCT WITH THE PROJECTIVE LINE: THE TWO FIBRES OF AN INTEGRAL CLOSED SUBSCHEME AND THE GENERATION OF RATIONALLY TRIVIAL CYCLES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The two fibres of a closed subscheme of the product as effective Cartier divisors. Divisors, label divisors-lemma-pullback-effective-Cartier-defined, invoked at proof line 3364. Gives part (1): because the generic point of W is not mapped into either of the two standard sections, the two fibres are effective Cartier divisors of W.
- The graph construction attached to a rational function on an integral scheme. Same chapter chow.tex, label lemma-rational-function, invoked at proof lines 3375 and 3397. Supplies both the rational equivalence of the two fibre classes on W for part (2) and the closed subscheme W_i inside V_i times the projective line whose two fibre classes realise the divisor of f_i in part (4).
- A proper morphism preserves rational equivalence. Same chapter chow.tex, label lemma-proper-pushforward-rational-equivalence, invoked at proof line 3377; the definition of rational equivalence is definition-rational-equivalence (used at proof lines 3355 and 3386). Moves the rational equivalence from W to X, since the projection is proper, which is what turns the equivalence on the graph into the statement about the fibres in X.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02OO: sources/stacks-chow/tags.txt, source lines 3523–3523. Navigation pointer for \ref{divisors-lemma-pullback-effective-Cartier-defined} used inside the counted span (line 3364); the argument that consumes it is retained in the span.

Source reference. Original lemma 02S2: sources/stacks-chow/pinned-original-source.tex, source lines 3217–3217. Navigation pointer for \ref{lemma-proper-pushforward-rational-equivalence} used inside the counted span (line 3377); the argument that consumes it is retained in the span.

Source reference. Original lemma 02RQ: sources/stacks-chow/pinned-original-source.tex, source lines 2701–2701. Navigation pointer for \ref{lemma-rational-function} used inside the counted span (line 3375); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 3336); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be a scheme locally of finite type over S. Let $W \subset X \times_S \mathbf{P}^1_S$ be an integral closed subscheme of δ-dimension $k+1$. Assume $W \neq W_0$, and $W \neq W_\infty$. Then*

1. *W_0, W_∞ are effective Cartier divisors of W,*
2. *W_0, W_∞ can be viewed as closed subschemes of X and*
$$[W_0]_k \sim_{rat} [W_\infty]_k,$$
3. *for any locally finite family of integral closed subschemes $W_i \subset X \times_S \mathbf{P}^1_S$ of δ-dimension $k+1$ with $W_i \neq (W_i)_0$ and $W_i \neq (W_i)_\infty$ we have $\sum([(W_i)_0]_k - [(W_i)_\infty]_k) \sim_{rat} 0$ on X, and*
4. *for any $\alpha \in Z_k(X)$ with $\alpha \sim_{rat} 0$ there exists a locally finite family of integral closed subschemes $W_i \subset X \times_S \mathbf{P}^1_S$ as above such that $\alpha = \sum([(W_i)_0]_k - [(W_i)_\infty]_k)$.*

Proof. Part (1) follows from Divisors, Lemma 02OO since the generic point of W is not mapped into D_0 or D_∞ under the projection $X \times_S \mathbf{P}^1_S \to \mathbf{P}^1_S$ by assumption.

Since $X \times_S D_0 \to X$ is a closed immersion, we see that W_0 is isomorphic to a closed subscheme of X. Similarly for W_∞. The morphism $p : W \to X$ is proper as a composition of the closed immersion $W \to X \times_S \mathbf{P}^1_S$ and the proper morphism $X \times_S \mathbf{P}^1_S \to X$. By Lemma 02RQ we have $[W_0]_k \sim_{rat} [W_\infty]_k$ as cycles on W. Hence part (2) follows from Lemma 02S2 as clearly $p_*[W_0]_k = [W_0]_k$ and similarly for W_∞.

The only content of statement (3) is, given parts (1) and (2), that the collection $\{(W_i)_0, (W_i)_\infty\}$ is a locally finite collection of closed subschemes of X. This is clear.

Suppose that $\alpha \sim_{rat} 0$. By definition this means there exist integral closed subschemes $V_i \subset X$ of δ-dimension $k+1$ and rational functions $f_i \in R(V_i)^*$ such that the family $\{V_i\}_{i \in I}$ is locally finite in X and such that $\alpha = \sum (V_i \to X)_* \mathrm{div}(f_i)$. Let
$$W_i \subset V_i \times_S \mathbf{P}^1_S \subset X \times_S \mathbf{P}^1_S$$
be the closure of the graph of the rational map f_i as in Lemma 02RQ. Then we have that $(V_i \to X)_* \mathrm{div}(f_i)$ is equal to $[(W_i)_0]_k - [(W_i)_\infty]_k$ by that same lemma. Hence the result is clear. □